

in down-time and ensuring quality for the customer.

He adds, "Latest technological innovation allows PLCs and vision systems to interact directly with the enterprise resource planning (ERP) layer, ensuring realistic Big Data processing to improve productivity and aid predictive maintenance, and much more. For example, our Sysmac NJ controller can read and write, to and from a SQL server at high speed. One practical application of this trend is serialisation or traceability, which is coming into the market."

A shift towards Industry 4.0

Perhaps the most recent and biggest trend in factory automation is the adoption of Industry 4.0. In fact, many experts feel that if India has to emerge as the preferred manufacturing destination, Industry 4.0 should be embraced with open arms.

A recent report by HIS technology forecasts that industrial automation will make up the majority of the Internet of Things (IoT) by 2025, and Internet Protocol (IP)-addressable devices are the key enabling forces that will get the industry there. The report notes, "Industrial automation accounted for more than half of the installed base for all Internet-connected devices in 2012. By 2025, the sector will account for nearly three-fourths of all connected devices—a CAGR of 36.3 per cent."

Dembi echoes a similar thought process, "The next industrial revolution, Industry 4.0, the Industrial IoT (IIoT) or whatever you want to call it, is one of the primary megatrends impacting our market today and will continue to do so in the foreseeable future." He opines that, today's concept of industrial automation is based on having one centralised brain collecting information from manufacturing assets to facilitate production decisions.

Industry 4.0 focuses on smart factories, connected machines, the IoT and the industrial Internet. Explaining how this works, Dembi



Industry 4.0 has been the talk of the town for quite some time now with the IoT and Big Data being some of the buzzwords associated with it. Not only is a connected factory a point of discussion but also easy diagnostic, quick maintenance and a seamless connectivity from the shop floor to the board." — Ninad Deshpande, specialist - open technologies, B&R Industrial Automation representing Ethernet POWERLINK Standardization Group



says, "In smart manufacturing, intelligence is decentralised, with each smartasset within the factory having full information about itself and equipped with the processing power to optimise its own productivity and efficiency. These smartassets are also hyper-connected to optimise and co-ordinate each step of the manufacturing process."

Explaining the application of such technology, Ninad Deshpande, specialist - open technologies, B&R Industrial Automation representing Ethernet POWERLINK Standardization Group, adds, "Interconnection of machinery, sensors and control systems together via intelligent networks helps us achieve a slew of features. These include dynamic response to product demands, enabling rapid manufacturing of new products, real-time optimisation of manufacturing, production and supply chain networks, highly-flexible production, strong customisation of products and mass customisation, self-optimisation, self-configuration and self-diagnostics."

He further stresses on the need of embracing open source communication in smartfactories to bring down the cost of ownership. He shares, "We as Ethernet POWERLINK Standardization Group focus on open source communications that help manufacturers get a solution for a connected world. We provide open source Ethernet POWERLINK technology that can be used by manufacturers for achieving real-time deterministic communication. This is completely licence-free, having no patents and copyrights."

Future outlook

Today, manufacturing has evolved to an extent where only through innovation and investment in technologies can businesses gain the productivity and efficiency advantage that will allow them to compete in the global market.

A Frost & Sullivan whitepaper titled 'Enhancing India's Manufacturing Competitiveness - Reality of Adopting Technologies and Trends,' concedes that the next industrial revolution will be based on unification of information among participants in the entire value chain, from product inception to design, manufacturing, services and even refurbishment. The report notes, "It will transform the manufacturing processes in sync with the speed of change in customer needs, which implies making the production process flexible without taking excess time."

The industrial automation will also evolve in sync with these market pulses. In fact, industrial automation is ahead of most other industries in the readiness for the IoT and, more specifically, for the IIoT. "When one looks at the deployment of sensors, actuators and low-level devices that are needed to enable the IIoT or the IIoT, industrial automation has an advantage," shares Dembi.

Keeping these trends in mind, Wadhwa feels that going forward, value-added operations in terms of engineering, software and testing will earn prominence, and the players that have got technical acumen in those areas are expected to grow at a faster rate. ●